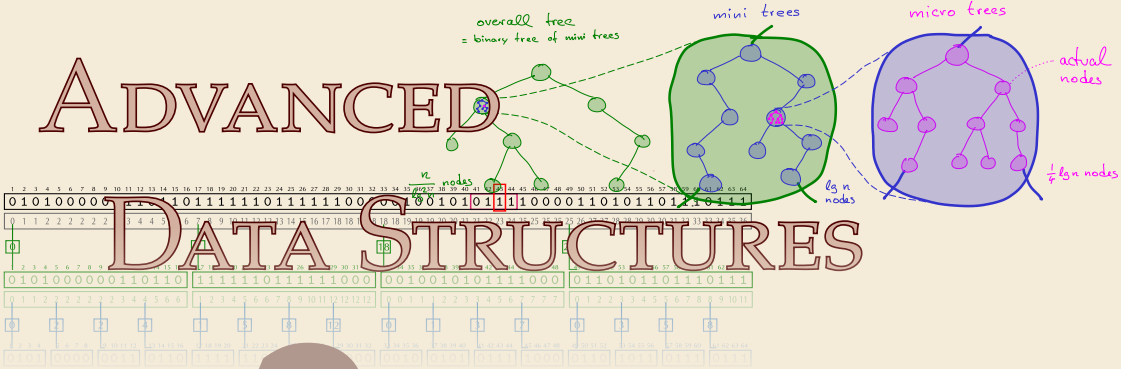
Advanced Data Structures



2 Adaptive Trees

Advanced Data Structures · Summer 2026

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CS650 (Summer 2026)
Philipps-Universität Marburg
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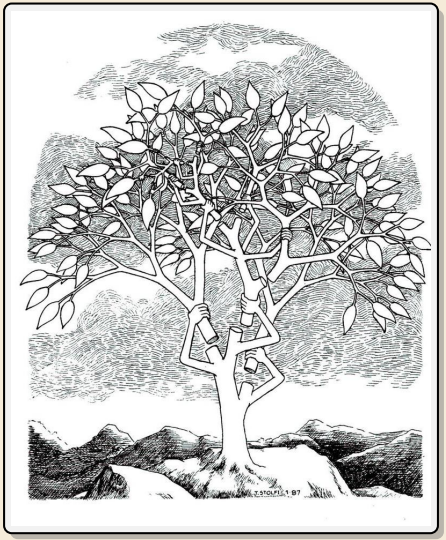
Outline

2 Adaptive Trees

2.1 Move-to-Root Heuristic
2.2 Move-to-Root Analysis
2.3 Splay Trees
2.4 Analysis of Splay Trees
2.5 Biased Search Trees
2.6 Dynamic Optimality
2.7 Lazy Search Trees

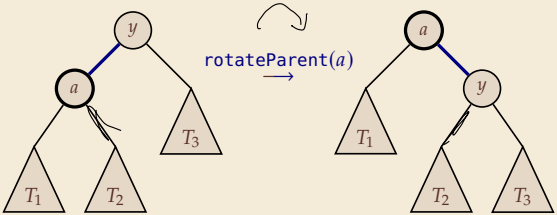
Self-Adjusting Data Structures

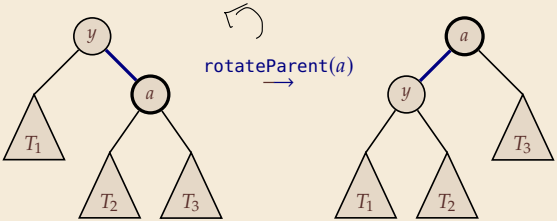
Idea: set of rules that makes data structure automatically adapt to use case



2.1 Move-to-Root Heuristic

Rotations


y
a
T1
T2
T3
rotateParent(a)
a
y
T1
T2
T3


y
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T1
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Move-to-root

Upon an access to an element, `rotateParent` until accessed element is at root!

Allen, Munro: *Self-Organizating Binary Search Trees*, JACM **1978**

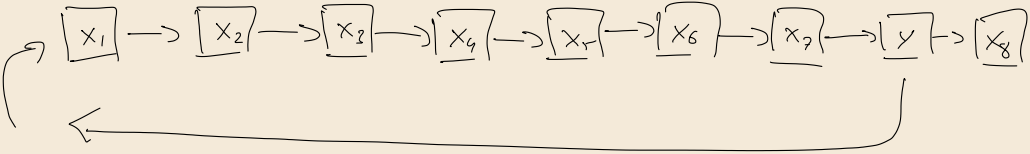
Inspired by **Move-To-Front heuristic** of self-adjusting linked **lists**

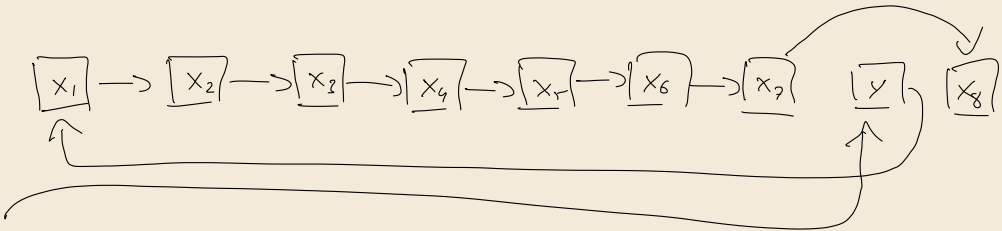
- cost via MTF is at most 2 times cost of optimal **static ordering**
- MTF is 2-competitive as an online algorithm
  (at most 2 times the cost of *any* algorithm)
- (both true under "basic cost model" only)
- with resource augmentation, LRU in paging very successful

Move-to-front

unsorted linked list

search(y)





Move-to-root

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Maybe, Move-to-root is similarly successful!

2.2 Move-to-Root Analysis

Iid Accesses

Iid Model: Suppose the accesses to keys $[1..n]$ in BST are *i.i.d. randomly* drawn with probabilities $p_1, \ldots, p_n$.

extreme case $\quad p_i = 2^{-i} \qquad p_n = 2^{-(n-1)}$

$\boxed{1} \to \boxed{2} \to \boxed{3} \to \boxed{4} \qquad (\underline{i})$

expected search cost $\quad \sum_{i=1}^{n} p_i (i-1) \approx \sum_{i=1}^{\infty} 2^i (i-1) = O(1)$

$$\sum_{i \geq 0} z^i = \frac{1}{1-z} \quad \Big| \frac{d}{dz}$$

$$\sum_{i \geq 0} i z^{i-1} = +1 \cdot (1-z)^{-2} = \frac{1}{(1-z)^2}$$

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$\rightsquigarrow$ Expected search cost in a BST T given by $\sum_{i=1}^{n} p_i \cdot \mathrm{depth}_T(i)$

Iid Accesses

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⇝ Expected search cost in a BST T given by $\sum_{i=1}^{n} p_i \left(\text{depth}_T(i) + 1\right)$

- If p_i known up front, can compute **static optimal BST** via dynamic programming

⇝ Cost of this optimal tree C_{OPT} can be bounded in terms of the entropy

$\left(p_i = 2^{-i} \quad \mathcal{H} = \sum 2^{-i} \cdot i \right)$

Lemma 2.1 (Bayer 1975)

$$\mathcal{H} - \lg \mathcal{H} - \lg e + 1 \;\leq\; C_{OPT} \;\leq\; \mathcal{H} + 1 \quad \text{for} \quad \mathcal{H} = \sum_{i=1}^{n} p_i \lg(1/p_i)$$

$0 \leq \mathcal{H} \leq \lg n$

Bayer: *Improved Bounds on the Costs of Optimal and Balanced Binary Search Trees*, M.Sc. Thesis, MIT **1975**

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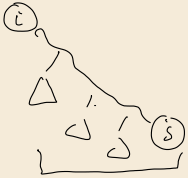
Note: C_{OPT} is for successful search / depth of internal nodes
The case of accesses to leaves only, with access to $\boxed{j}$ w/p q_j, $j = 0, \ldots, n$ even cleaner:

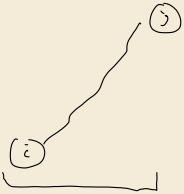
$$\mathcal{H} \;\leq\; \sum_{j=0}^{n} q_j \cdot \mathrm{depth}_T(\boxed{j}) \;\leq\; \mathcal{H} + 2$$

Move-to-Root on iid Accesses

Lemma 2.2 (MTR ancestors)

Let T be a tree maintained by the move-to-root heuristic and $i < j$ two keys.
Then i is an **ancestor** of j iff the most recent request for i came after the most recent requests of any of $i+1, \ldots, j$.
Similarly, j ancestor of i iff j requested more recently than any of $i, \ldots, j-1$.





Proof: "$\Rightarrow$" i ancestor of j cannot have accessed (MTR) $i+1, \ldots, j$ after i

"$\Leftarrow$" accesses to all of $i, \ldots, j$ but i last

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Recall $A^i_j = [i \text{ ancestor of } j]$

Corollary 2.3

Under the iid accesses model and MTR heuristic, $\mathbb{P}[A^i_j] = \begin{cases} \dfrac{p_i}{p_i + \cdots + p_j} & i < j \\ \dfrac{p_i}{p_j + \cdots + p_i} & i > j \end{cases}$

Move-to-Root on iid Accesses – Analysis

Theorem 2.4 (MTR cost)

Always assume: every key accessed at least once

Let C_{MTR} be the expected search cost under the i.i.d. model in a tree T maintained by MTR.

$$C_{MTR} = 1 + 2 \sum_{1 \le i < j \le n} \frac{p_i p_j}{p_i + \cdots + p_j} \le 1 + 2 \underset{\substack{\shortparallel \\ 1.99}}{\ln 2} \cdot \mathcal{H}$$

$$\mathcal{H} = \sum_{i=1}^{n} p_i \lg\left(\frac{1}{p_i}\right)$$

Move-to-Root has close to optimal search cost in the i.i.d. model!

Move-to-Root on iid Accesses – Analysis

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Proof:

$$C_{MTR} \;=\; \sum_{i=1}^{n} p_i \cdot (\overset{\#\text{ comps}}{\text{depth}(i) + 1})$$

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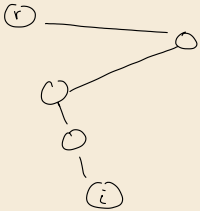

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Proof:

$$C_{MTR} \;=\; \sum_{i=1}^{n} p_i \cdot \mathrm{depth}(i) \;=\; \sum_{i=1}^{n} p_i \cdot \left(1 + \sum_{j \ne i} \mathbb{E}[A_i^j] \right)$$

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$$= \; 1 \; + \; \sum_{i=1}^{n} \sum_{j \ne i} p_i \mathbb{E}[A_i^j]$$

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$$= \; 1 + \sum_{i=1}^{n} \sum_{j \ne i} p_i \mathbb{E}[A_i^j] \;\underset{\text{Lemma}}{=}\; \frac{p_j}{p_i + \cdots + p_j}$$

$$= \; 1 + 2 \sum_{1 \le i < j \le n} \frac{p_i p_j}{p_i + \cdots + p_j}$$

This proves the first part.

Move-to-Root on iid Accesses – Analysis

Proof (cont.):

Mehlhorn's trick: $\delta_{i,j} := \dfrac{p_i + \cdots + p_j}{p_i} \quad \rightsquigarrow \quad 1 = \delta_{i,i} \leq \delta_{i,i+1} \leq \cdots \leq \delta_{i,n} \leq \dfrac{1}{p_i}$

Move-to-Root on iid Accesses – Analysis

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$$C_{MTR} = 1 + 2\sum_{i=1}^{n-1} p_i \sum_{j=i+1}^{n} \frac{p_j}{p_i + \cdots + p_j}$$

Move-to-Root on iid Accesses – Analysis

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$$\frac{\delta_{i,j} - \delta_{i,j-1}}{\delta_{i,j}}$$

$$= \frac{\dfrac{\cancel{p_i + \cdots + p_{j-1}} + p_j - \cancel{(p_i + \cdots + p_{j-1})}}{\cancel{p_i}}}{\dfrac{p_i + \cdots + p_j}{\cancel{p_i}}}$$

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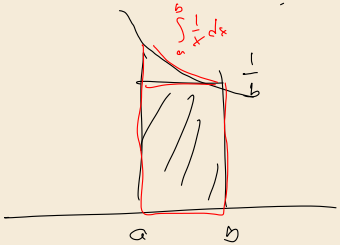
$$\boxed{a \le b \implies (b-a)\cdot\tfrac{1}{b} = \int_a^b \tfrac{1}{b}\,dx < \int_a^b \tfrac{1}{x}\,dx}$$



Move-to-Root on iid Accesses – Analysis

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\begin{aligned}
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&= 1 + 2\sum_{i=1}^{n-1} p_i \sum_{j=i+1}^{n} \frac{\delta_{i,j} - \delta_{i,j-1}}{\delta_{i,j}} \\
&< 1 + 2\sum_{i=1}^{n-1} p_i \sum_{j=i+1}^{n} \int_{x=\delta_{i,j-1}}^{\delta_{i,j}} \frac{1}{x} dx
\end{aligned}
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Move-to-Root on iid Accesses – Analysis

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$$\boxed{\int_{x=a}^{b} \tfrac{1}{x}\,dx = \ln(b) - \ln(a) = \ln(b/a)}$$

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&= 1 + 2\sum_{i=1}^{n-1} p_i \ln(\delta_{i,n}/\delta_{i,i}) \le 1 + 2\ln 2 \sum_{i=1}^{n-1} p_i \lg(1/p_i) = 1 + 2\ln 2 \cdot \mathcal{H}
\end{aligned}$$

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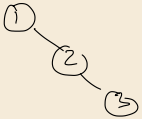
■

Unbalanced BSTs under i.i.d. model

Interestingly, the same cost result from unbalanced BSTs!

- Suppose we repeatedly draw x i.i.d. from $[1..n]$ w/p $p_1, \ldots, p_n$
- We insert x into initially empty unbalanced BST T
- **If x is already contained in T, the insertion has no effect**
- Repeat until saturation (i.e., until we have seen every value $x \in [1..n]$)

$n=3$ 1 2 2 1 3 2 1 2



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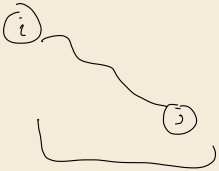

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$$\rightsquigarrow C_T = 1 + 2 \sum_{1 \le i < j \le n} \frac{p_i p_j}{p_i + \cdots + p_j} \le 1 + 2 \ln 2 \cdot \mathcal{H}$$

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- Repeat until saturation (i.e., until we have seen every value $x \in [1..n]$)

Since insertions are i.i.d., probability that i is inserted first among $[i..j]$ is $\dfrac{p_i}{p_i + \cdots + p_j}$

$$\rightsquigarrow \mathbb{P}[A_j^i] = \begin{cases} \dfrac{p_i}{p_i + \cdots + p_j} & i < j \\ \dfrac{p_i}{p_j + \cdots + p_i} & i > j \end{cases}$$

$$\rightsquigarrow C_T = 1 + 2 \sum_{1 \le i < j \le n} \frac{p_i p_j}{p_i + \cdots + p_j} \le 1 + 2 \ln 2 \cdot \mathcal{H}$$

Both unbalanced BST and Move-to-Root behave well under i.i.d. model.

Move-to-Root's Blind Spot

Problem: *Can't rely on accesses to be random.*

Theorem 2.5 (Move-to-Root Bad Case)

There exists a sequence of n requests, such that starting from any tree T, the amortized access cost over the course of the n requests is $\Omega(n)$.

Proof:
see exercises

2.3 Splay Trees

Splay Trees

Splay trees by now the most widely known self-adjusting data structure.

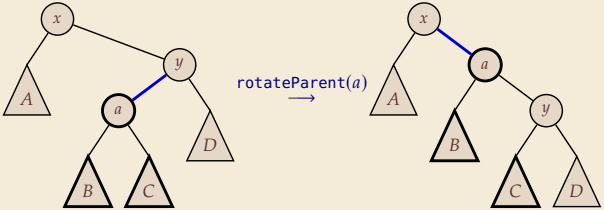
- 👍 reasonably simple and fast
- 👍 simple to implement
- 👍 remarkable theoretical properties (*access theorems*) ⇝ later
- 👎 many writes and pointer chases

Sleater, Tarjan: *Self-Adjusting Binary Search Trees*, JACM **1985**

Move-to-root reloaded

What's wrong in Move-to-front?

Let's look at the possible cases of parent **and grandparent**



Move-to-root reloaded

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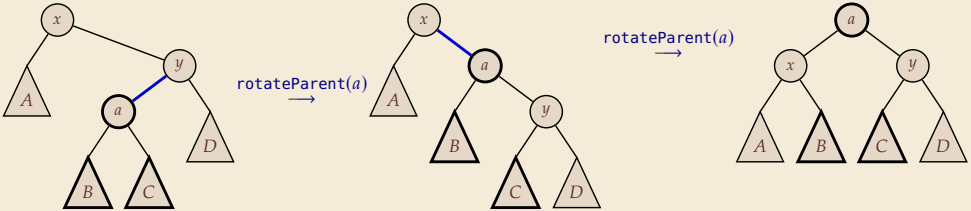

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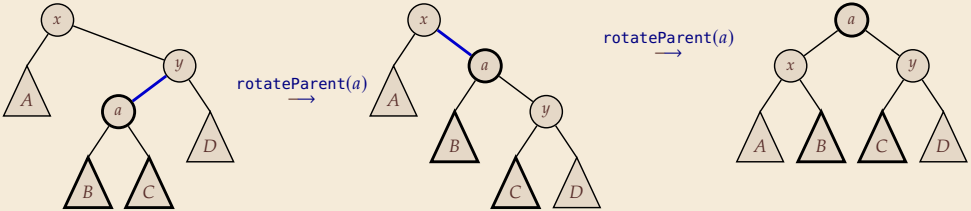

Using simple `rotateParent` calls, the subtree we came from (B or C) is lifted up!

Move-to-root reloaded

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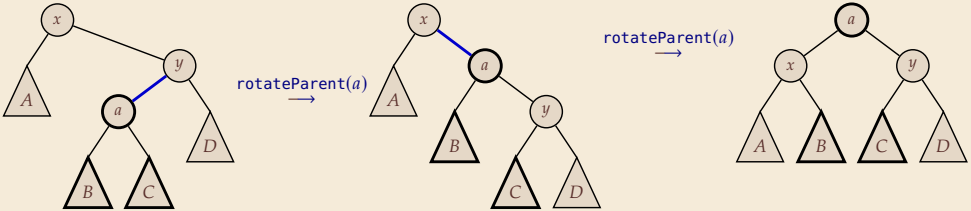

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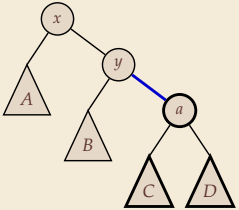

Move-to-root reloaded

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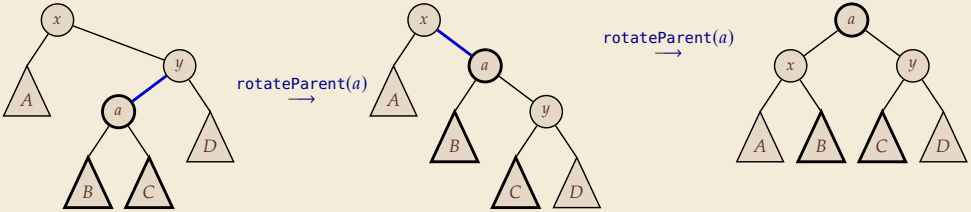

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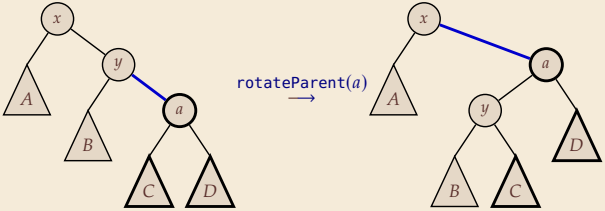

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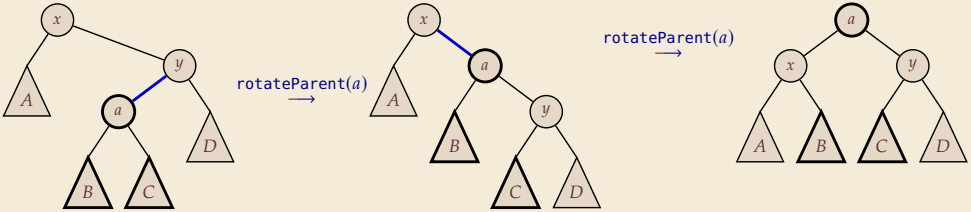

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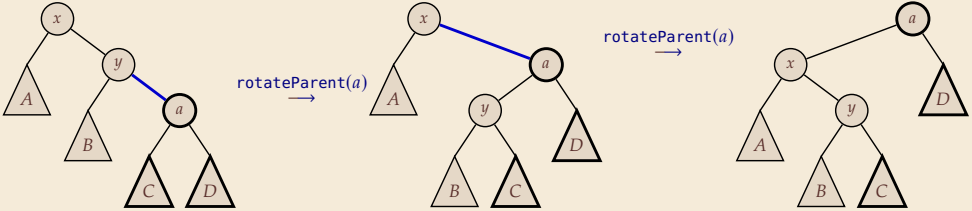

Move-to-root reloaded

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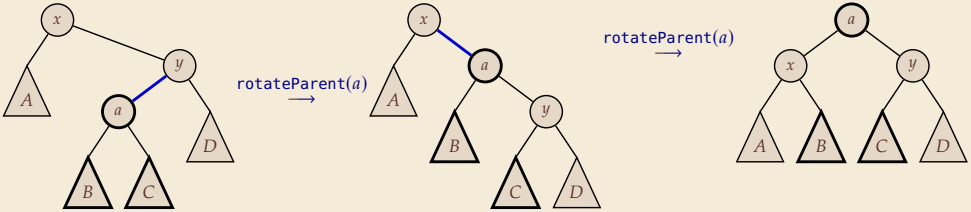

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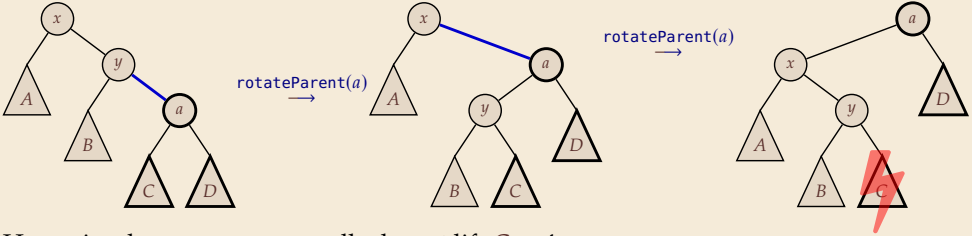
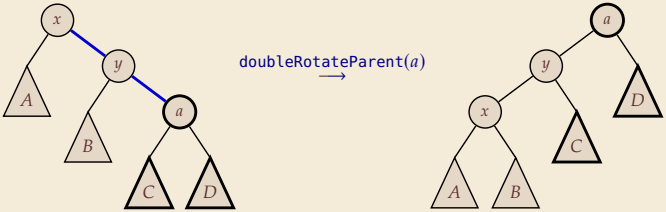


Here, simple `rotateParent` calls do **not** lift C up!

The Fix

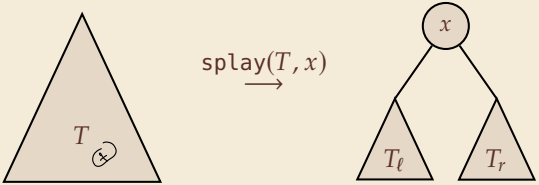
In the case that both parents are right children, do a double rotation



Now both C and D are lifted up!

Splay

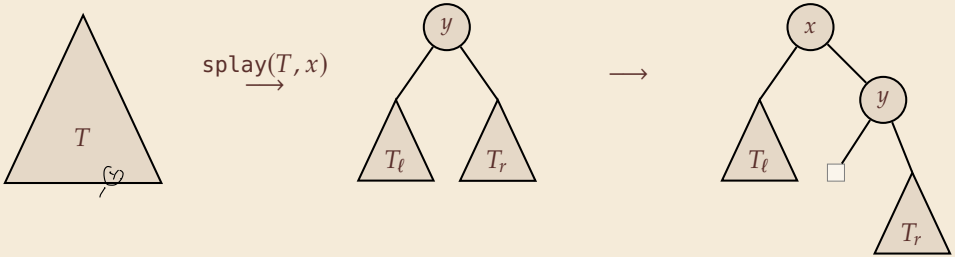
Aggressive restructuring allows to base all operations on single primitive: `splay`




```
def splay(T, x):
    while x ≠ T.root
        p = parent(x)
        if p == T.root
            rotateParent(x)  # Zig Case
        else:
            g = parent(p)
            if p == g.left ∧ x == p.left or p == g.right ∧ x == p.right:
                doubleRotateParent(x)  # Zig-Zig Case
            else:
                rotateParent(x)  # Zig-Zag Case
                rotateParent(x)
```

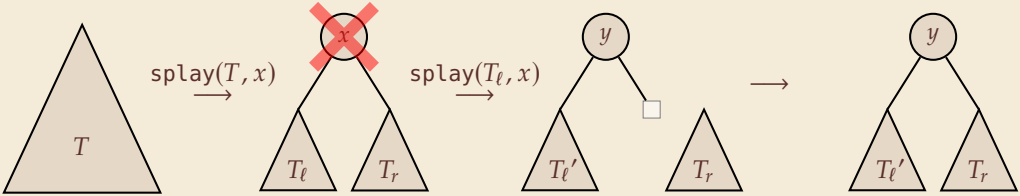
Splay – Insert

1. `splay`(x, T) $\rightsquigarrow$ root y and $T_\ell \leq y \leq T_r$ (key order)
2. If $x = y$ (key already stored) return.
3. Otherwise, w.l.o.g. $x < y$, can simply make x new root.



Splay – Delete

1. $\texttt{splay}(x, T) \rightsquigarrow$ root x (or not part of tree and we're done)
2. Remove x, left with T_ℓ and T_r
3. $\texttt{splay}(x, T_\ell) \rightsquigarrow$ root $y = \texttt{max}(T_\ell)$ and $y.right = \texttt{null}$
4. $y.right := T_r$



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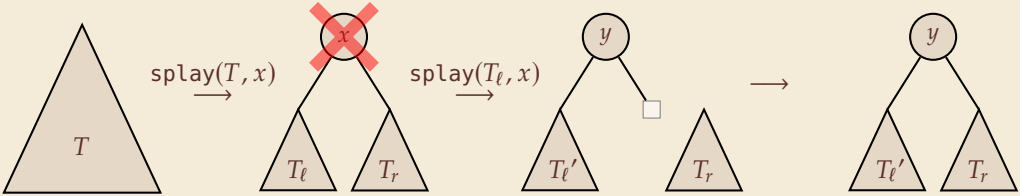

Note: We've effectively built **`join`** $\rightsquigarrow$ naturally supported via `splay`.

`split` even more directly supported via `splay`: $\texttt{split}(T, x)$: $\texttt{splay}(T, x)$, return $(x.left, x)$

https://www.cs.usfca.edu/~galles/visualization/SplayTree.html

2.4 Analysis of Splay Trees

Rotation Cost

Note: In all operations, we follow a search path and splay.
(For delete twice)

splay walks back up search path

- $\rightsquigarrow$ use $k = \Theta(d)$ rotations for splaying node at depth d.
- $\rightsquigarrow$ cost is dominated by k = **#rotations in `splay`** calls.

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Clearly, individual operations could be expensive (use $k = \Omega(n)$ rotations).

$\rightsquigarrow$ Can only hope for an ***amortized*** bound on k

Idea of amortized analysis : sequence of operations $op_1, op_2, op_3, \ldots, op_m$

c_i = actual cost of op_i (chaotic) $\qquad \sum_{i=1}^{m} c_i \leq B(n)$

a_i = amortized cost of op_i := $c_i + \Delta\Phi_i$

Φ = "potential" state of ds $\to \mathbb{R}$ $\qquad \Phi_i - \Phi_{i-1}$

Φ_i = potential after op_i $\quad$ Φ_0 initial potential

$$\sum_{i=1}^{m} a_i = \sum_{i=1}^{m} \left(c_i + \Phi_i - \Phi_{i-1} \right) = \sum_{i=1}^{m} c_i + \Phi_m - \Phi_0$$

$$\sum_{i=1}^{m} c_i = \sum_{i=1}^{m} \underbrace{a_i}_{\leq B(n)} + \Phi_0 - \Phi_m \leq m \cdot B(n) + \Phi_0 - \Phi_m$$

Rotation Cost

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Clearly, individual operations could be expensive (use $k = \Omega(n)$ rotations).

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We define a potential function $\Phi = \Phi(T)$ such that

$\Delta\Phi$ = change in potential

Goal: If `splay`(x) uses k rotations, then $\boxed{k + \Phi' - \Phi \quad \leq \quad 1 + 3 \cdot \big(r'(x) - r(x)\big)}$

change in "rank" of x

$\rightsquigarrow$ Released potential can pay for actual cost k